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The effects of riverine impoundment on genetic structure and gene flow in two stream fishes in the Mobile River basin

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SUMMARY

1. Riverine impoundments (reservoirs) are thought to impede natural migration in small-stream-inhabiting fishes, resulting in spatially and genetically fragmented populations. However, this hypothesis remains poorly tested, and the genetic consequences of riverine impoundment for stream fishes with differing dispersal capabilities are not well understood.
2. This study utilised a combination of microsatellite DNA loci from 479 individuals and the mitochondrial (mt) DNA cytochrome *b* gene (810–1140 bp) from 83 individuals from eight streams to compare genetic structure and diversity between reservoir-fragmented and non-fragmented groups of two species of stream fishes with differing dispersal capabilities in the Mobile River basin of the south-eastern United States.
3. For both species, analysis of microsatellite loci revealed genetic discontinuities between neighbouring tributaries that have been fragmented by reservoir construction. This finding was in stark contrast to the high degree of continuity for both species between reference tributaries in a natural river setting. Results from mtDNA revealed no significant genetic structure within reservoir-fragmented or non-fragmented groups, indicating a lack of historical genetic structure (i.e. prior to reservoir construction). Microsatellite-based estimates of genetic diversity and migration were differentially affected in the two species, indicating that stream fishes with relatively high dispersal abilities may be equally or more susceptible to reservoir fragmentation when compared to species with relatively low dispersal abilities.
4. Collectively, our data revealed that riverine impoundment and reservoir-induced habitat fragmentation adversely affect genetic characteristics in small-stream-inhabiting fishes. This is of particular interest in biodiversity hotspots such as the south-eastern United States where hydroelectric and recreational reservoirs restrict connectivity in aquatic systems.

Keywords: dams, genetic structure, habitat fragmentation, microsatellites, reservoirs

Introduction

Habitat fragmentation is one of the leading causes of decline in biodiversity worldwide (Fahrig, 2003; Fischer & Lindenmayer, 2007; Lindenmayer *et al*., 2008; Heller & Zavaleta, 2009). Anthropogenic habitat alteration often results in the fragmentation of previously continuous migration routes, resources or environmental conditions (Segelbacher *et al*., 2010). For many organisms, habitat fragmentation and overall loss of favourable habitat leads to subdivision into smaller, isolated populations with reduced or no connectivity (Kindlmann & Burel, 2008). Of particular interest for the conservation of biodiversity is the degree to which contemporary habitat discontinuity affects functional connectivity via individual exchange and gene flow (Fischer & Lindenmayer, 2007). Reduced gene flow among populations leads to reduced intra-population genetic variation, which can

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affect fitness, population persistence and adaptation by populations to changing environmental conditions (Lande, 1987; Reed & Frankham, 2003; DiBattista, 2008).

Over the past century, the construction of dams and their associated riverine impoundments has decreased connectivity in freshwater systems worldwide, raising major ecological concern (e.g. Benke, 1990; Nilsson & Berggren, 2000; Pringle, Freeman & Freeman, 2000; Richter *et al.*, 2003; Xie, 2003). The negative impacts of riverine impoundment on migration and genetic connectivity in large-bodied or migratory fishes are well documented (see Fullerton *et al.*, 2010). However, a major proportion of the ichthyofauna in most freshwater systems consists of small-stream-inhabiting fishes (hereafter referred to as stream fishes) (Etnier, 1997; Lundberg *et al.*, 2000). Only recently have studies revealed the potentially negative impacts of riverine impoundments on genetic connectivity in stream fishes, highlighting the need for more comprehensive studies to understand how these impoundments affect this large proportion of aquatic biodiversity (Skalski *et al.*, 2008; Franssen, 2011; Hudman & Gido, 2013).

For stream fishes, dams can pose barriers to intra- and inter-drainage migration and gene flow in two main ways. A dam may simply be a physical barrier to passage that impedes upstream or downstream migration and gene flow (or both) between stream fishes in neighbouring streams (Raeymaekers *et al.*, 2009; Junker *et al.*, 2012). Migration and gene flow in stream fishes may also be blocked by riverine impoundments created by the dam, which eliminates stretches of preferable stream habitat and creates large expanses of inhospitable habitat between neighbouring streams (Herbert & Gelwick, 2003; Skalski *et al.*, 2008; Franssen, 2011; Hudman & Gido, 2013). For example, reservoir pools may inundate large portions of previously free-flowing stream habitat and sever lotic corridors between neighbouring streams. This type of reservoir-induced fragmentation has been shown to negatively affect genetic characteristics in populations of a stream specialist fish (creek chub, *Semotilus atromaculatus*; Skalski *et al.*, 2008; Hudman & Gido, 2013) and may affect gene flow between populations of habitat generalist species such as the red shiner, *Cyprinella lutrensis* (Franssen, 2011).

The fresh waters of eastern North America contain one of the world's most diverse fish faunas, where biodiversity loss is strongly associated with alteration of freshwater habitats (Warren *et al.*, 1997, 2000; Ricciardi & Rasmussen, 1999; Jelks *et al.*, 2008). The influence of riverine impoundment on aquatic biodiversity loss is exemplified in the species-rich Mobile basin (Fig. 1a), where 34 major hydroelectric and navigation dams impound approximately 44% of river mainstem and extensively fragment lotic habitats (Burke & Ramsey, 1995; Pringle *et al.*, 2000). Dam construction and hydrological alteration in the Mobile basin are among the main explanations for decline of the native fish fauna, where approximately 30% of more than 190 native fish species are considered endangered, threatened or vulnerable (Etnier, 1997; Warren *et al.*, 1997; Jelks *et al.*, 2008). In addition, at least 16 endemic mussel species and approximately 32% of the native gastropod fauna are presumed extinct as a result of riverine impoundments in the Mobile basin (Lydeard & Mayden, 1995; Neves *et al.*, 1997; Pringle *et al.*, 2000). Thus, the Mobile basin offers a natural laboratory for studying the influence of riverine impoundment on connectivity in stream fishes.

The objective of this study was to determine how reservoir-induced fragmentation influences genetic structure, gene flow and genetic diversity in stream fishes in a Mobile basin river system. Using a combination of microsatellite and mitochondrial (mt) DNA loci and population genetic methods, we examined genetic structure and diversity between reservoir-fragmented and non-fragmented groups of two species of stream fishes with differing dispersal capabilities in the Tallapoosa River system. Our results contribute to the understanding of impacts of riverine impoundment on aquatic biodiversity in the Mobile basin and highlight the utility of the Mobile basin as a model system to better predict biodiversity loss in other impounded rivers. Given the old age and large size of Martin Lake and its relatively small extant tributary segments, we hypothesise that, for both species, the reservoir group will have higher levels of inter-tributary genetic structure and lower levels of intra-tributary genetic diversity compared with the river group. Alternatively, if reservoir fragmentation does not impede migration and gene flow in stream fishes, there will be no difference in genetic structure and diversity between reservoir and river groups for both species. We further predict that species with low dispersal capabilities will be more susceptible to the negative impacts of reservoir fragmentation. If this prediction is true, the degree of genetic structuring and loss of genetic diversity will be greater between reservoir and river groups of the relatively sessile *Etheostoma tallapoosae* when compared to reservoir and river groups of the relatively mobile *Cyprinella gibbsi*. Alternatively, if there is no difference in the degree of genetic differentiation and variability between reservoir and river groups of both species, then differences in dispersal ability may not play a major role in the ability of a species to overcome barriers to gene flow

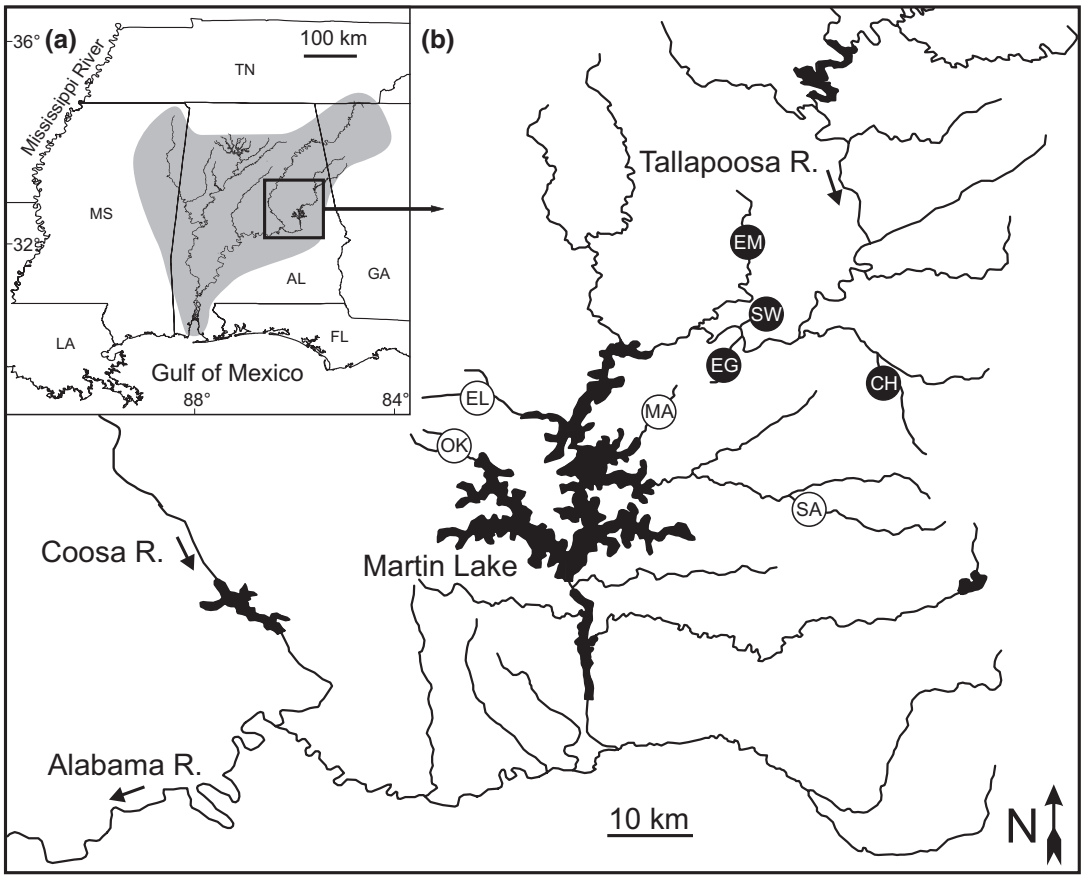

Fig. 1 (a) Map of the Mobile River basin (shaded), highlighting the sampling area within the Tallapoosa River drainage. (b) Tributaries sampled for *Cyprinella gibbsi* and *Etheostoma tallapoosae* used in this study. Reservoir tributaries (white circles) sampled: (OK) Oakachoy Creek, (SA) Sandy Creek, (MA) Manoy Creek, (EL) Elkahatchee Creek. River tributaries (black circles) sampled: (EM) Emuckfaw Creek, (SW) Sweetwater Creek, (EG) Eagle Creek, (CH) Little Chatahospee Creek.

in a reservoir-fragmented setting. We further evaluate demographic history in reservoir-fragmented versus non-fragmented settings and discuss differences with respect to current patterns of genetic structure.

Methods

Study system

The Tallapoosa River was selected because of the high number of endemic fish species (Warren *et al*., 1997) and the presence of large lengths of both impounded and unimpounded river mainstem. Our fragmented group (referred to hereafter as the 'reservoir group') consisted of four streams that enter Martin Lake, a riverine impoundment of the mainstem of the Lower Tallapoosa River (Fig. 1b). Martin Lake was constructed in 1926 and is the largest reservoir in the Mobile basin, covering 16 187 hectares (Burke & Ramsey, 1995). Our non-fragmented group (referred to hereafter as the 'river group') consisted of four streams that enter the unimpounded reach of the Tallapoosa River upstream of Martin Lake (Fig. 1b). The river group was selected to represent natural inter-tributary connectivity within the Tallapoosa River (i.e. their connectivity is putatively unimpeded by impoundment effects). As suggested by Segelbacher *et al*. (2010), landscape genetic studies treating multiple landscapes as independent observations are best suited to rigorously test the pattern–process relationship of landscape effects. Thus, in all genetic analyses, we treated the two differing landscapes (i.e. reservoir and river groups) as independent observations.

The tallapoosa shiner (*C. gibbsi*; Fig. 2a) is a small [maximum 82 mm standard length (SL)] mid-water dwelling minnow endemic to the Tallapoosa River drainage, occurring in small- to medium-sized streams (Howell & Williams, 1971; Boschung & Mayden, 2004). The tallapoosa darter (*E. tallapoosae*, subgenus *Ulocentra*; Fig. 2b) is a small (maximum 64 mm SL) benthic darter that co-occurs with *C. gibbsi* in small- to medium-sized streams (Boschung &

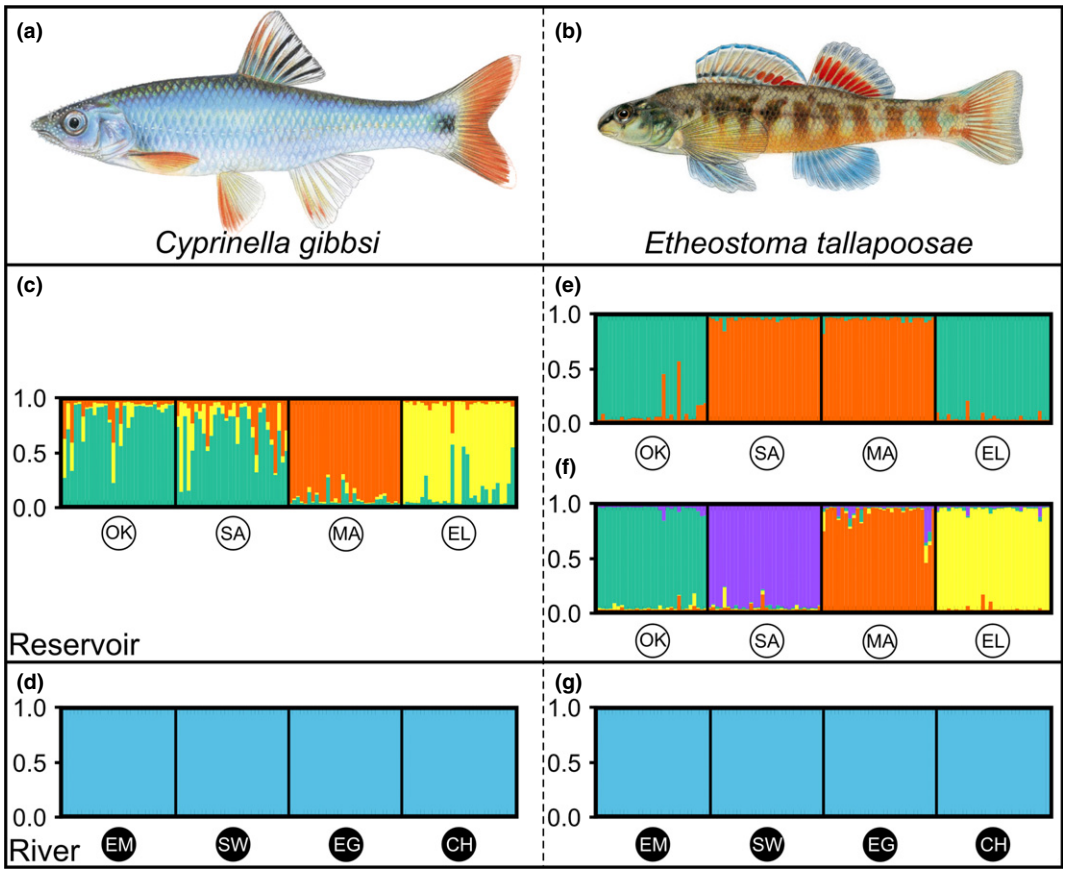

Fig. 2 Bar plots depicting results from the microsatellite-based STRUCTURE analyses for (a) tallapoosa shiner, *Cyprinella gibbsi*, and (b) tallapoosa darter, *Etheostoma tallapoosae*. Bars correspond to multilocus genotypes of individuals, and colours represent the probability of ancestry to each STRUCTURE inferred cluster (*K*). (c) Most likely genetic structure ($K = 3$) for the reservoir group of *C. gibbsi*. (d) Most likely genetic structure ($K = 1$) for the river group of *C. gibbsi*. (e) Bar plot of $K = 2$ for the reservoir group of *E. tallapoosae*. (f) Bar plot of $K = 4$ for the reservoir group of *E. tallapoosae*. (g) Most likely genetic structure ($K = 1$) for the river group of *E. tallapoosae*.

Mayden, 2004). Although aspects of dispersal ability of *C. gibbsi* are not well known, Boschung & Mayden (2004) suggest biological similarities with other species of *Cyprinella*, which can be highly mobile (Walters *et al.*, 2008). Conversely, members of the darter subgenus *Ulocentra* have been shown to exhibit low dispersal abilities and remarkable patterns of population structuring within river drainages (Clabaugh *et al.*, 1996; Porter, Cavender & Fuerst, 2002). Collection records indicate that both species specialise in habitats of bedrock, gravel and sand in small-medium flowing streams, with occasional collections from the free-flowing portions of the Tallapoosa River mainstem with appropriate habitat, and no known collections from the reservoir pool of Martin Lake (Howell & Williams, 1971; Suttkus & Etnier, 1991; Boschung & Mayden, 2004). These distributional patterns, in addition to the lack of appropriate substratum and spawning habitat in reservoirs, indicate that lentic habitats of reservoirs probably represent inhospitable conditions for these two species. Previous studies of mtDNA variation in both species revealed slightly higher levels of genetic structuring in *E. tallapoosae* when compared to *C. gibbsi*, and although both species exhibited some rangewide genetic structuring, both showed evidence for recent or ongoing gene flow between populations separated by approximately 90 km in unimpounded sections of the Tallapoosa River (Brogdon, Tabit & Kral, 2003; Connelly, Tabit & Kral, 2006).

Specimen collection and compatibility of landscapes

The reservoir group consisted of samples from two western tributaries [Oakachoy Creek (OK) and Elkahatchee Creek (EL)] and two eastern tributaries [Sandy Creek (SA) and Manoy Creek (MA)] that flow into Martin Lake (Fig. 1b; Table 1). The river group consisted of samples from two western tributaries [Emuckfaw Creek (EM) and Sweetwater Creek (SW)] and two eastern tributaries [Eagle Creek (EG) and Little Chatahospee Creek (CH)] that flow into the Tallapoosa River mainstem upstream of reservoir impoundment (Fig. 1b; Table 1). To evaluate

the compatibility of our sampling strategy between the two landscapes, nonparametric statistics (Mann–Whitney U-tests) were used to test for potential differences in distance between stream mouths and for distance between sampling locations between the two landscapes. Geographical distances were measured with the program DIVA-GIS (Hijmans *et al.*, 2004) as the shortest distance by water in kilometres, using the most downstream sample locality in each tributary. A total of 479 individuals were collected by seine comprising 240 *C. gibbsi* and 239 *E. tallapoosae* (Table 1). For all fishes, a small fin clip was removed and preserved in 95% ethanol, and genomic DNA was extracted using the DNEasy Kit (Qiagen, Valencia, CA, U.S.A.). Specimens were preserved in formalin and catalogued into the University of Alabama Ichthyological Collection (UAIC catalog numbers listed in Table 1).

Microsatellite genotyping

All *C. gibbsi* were genotyped for ten microsatellite loci using primers designed for *Campostoma anomalum* (Can6EPA; Dimsoski, Toth & Bagley, 2000), *Cyprinella lutrensis* (Cylu04_01_H06; Monnahan *et al.*, 2009), *Luxilus cornutus* (Lco1, Lco5; Turner *et al.*, 2004), *Notropis mekistocholas* (Nme24B6.191, Nme24B6.211, Nme18C2.178, Nme25C8.208; Burridge & Gold, 2003) and *Rhinichthys cataractae* (Rhca20, Rhca24; Girard & Angers, 2006). All *E. tallapoosae* were genotyped for ten microsatellite loci using primers designed for *E. caeruleum* (Eca10EPA, Eca48EPA, Eca49EPA; Tonnis, 2006), *E. osburni* (EosC6, EosC112; Switzer, Welsh & King, 2008) and *E. scotti* (Esc18, Esc26b, Esc57, Esc68, Esc132b; Gabel *et al.*, 2008). Microsatellite DNA fragments were amplified for each sample via polymerase chain reaction (PCR) using the GoTaq* Flexi DNA polymerase kit (Promega Corp., Madison, WI, U.S.A.) and run conditions of Fluker *et al.* (2010). Fluorescently labelled PCR fragments (HEX and 6-FAM) were mixed with GeneScan™ 500 ROX™ Size Standard (Applied Biosystems, San Francisco, CA, U.S.A.) and read on an ABI 3730 (Applied Biosystems) DNA analyzer (University of Maine DNA sequencing facility).

Characteristics and genetic diversity of microsatellite DNA

Prior to analyses, alleles were binned using the program FLEXIBIN 2.0 (Amos *et al.*, 2007). The resulting data sets were then screened for typographic errors and genotyping

Table 1 Sampling locations, sample sizes, and University of Alabama Ichthyological Collection (UAIC) catalogue numbers for tallapoosa shiners (*Cyprinella gibbsi*) and tallapoosa darters (*Etheostoma tallapoosae*) used in this study. Symbols correspond to Fig. 1.

Group/Tributary/Locality	Symbol	Latitude	Longitude	*C. gibbsi* (*N*, *N* mtDNA)	*E. tallapoosae* (*N*, *N* mtDNA)	UAIC catalogue number
Reservoir						
Oakachoy Creek	OK					
Co. Rd. 26		32.84861	−86.04667	5, 3	29, 3	15741.01, 15741.02
AL Hwy 259		32.83389	−86.04028	25, 3	0, 2	15740.01, 15741.03
Sandy Creek	SA					
Co. Rd. 89		32.78028	−85.65306	30, 5	2, 2	15742.01, 15742.02
Co. Rd. 011		32.75444	−85.57278	0, 0	19, 2	15743.02
Tributary at Co. Rd. 002		32.76306	−85.53333	0, 0	9, 2	15744.01
Manoy Creek	MA					
US Hwy 280		32.87333	−85.80917	30, 5	30, 5	15745.01, 15745.02
Elkahatchee Creek	EL					
AL Hwy 22		32.90500	−86.01028	30, 5	30, 5	15739.01, 15739.03
River						
Emuckfaw Creek	EM					
Co. Rd. 98		33.07889	−85.69472	30, 5	30, 5	15735.01, 15735.04
Sweetwater Creek	SW					
Sweetwater Rd.		32.99184	−85.69430	30, 5	30, 5	15754.01, 15754.03
Eagle Creek	EG					
Duck Rd.		32.92913	−85.73120	0, 0	19, 3	15748.01
Russell Ida Rd.		32.95336	−85.71649	30, 5	11, 3	15749.01, 15749.02
Little Chatahospee Creek	CH					
Tributary at Co. Rd. 32		32.85694	−85.50333	0, 0	30, 5	15756.05
Co. Rd. 66		32.92917	−85.53056	30, 5	0, 0	15757.01, 15758.01

N, number of individuals genotyped for microsatellite DNA loci; *N* mtDNA, number of individuals sequenced for the mitochondrial DNA cytochrome *b* gene.

errors due to non-amplified alleles (null alleles) and/or short allele dominance (large allele dropout) using the program MICRO-CHECKER (van Oosterhout *et al.*, 2004). Deviations from Hardy–Weinberg equilibrium (HWE) expectations were assessed with exact tests in GENEPOP version 4.0.10 (Rousset, 2008) using a Markov chain algorithm with 10 000 dememorisations, 200 batches and 10 000 iterations per batch. Linkage disequilibrium (LD) was tested across all pairs of loci using 10 000 dememorisations, 500 batches and 5000 iterations per batch in GENEPOP. The program ARLEQUIN v3.5 (Excoffier & Lischer, 2010) was used to calculate the mean number of alleles per locus (A), observed heterozygosity (H_o) and unbiased expected heterozygosity (H_e; Nei, 1978). We also calculated allelic richness (AR, a measure of allelic variation that accounts for differences in sample sizes and missing data) and private alleles using the program HP-RARE (Kalinowski, 2005). The program FSTAT v 2.9.3.2 (Goudet, 2001) was used to test the hypothesis that genetic diversity (based on AR, H_o, and H_e) is lower in the reservoir group compared with the river group for both species.

Amplification and sequencing of mtDNA

A subsample of individuals from each species was sequenced for the mtDNA cytochrome *b* (cyt *b*) gene (Table 1), and sequences were deposited in GenBank under accession numbers KF058909–KF058933 (see Appendix S1 in Supporting information for specific haplotype designations and corresponding GenBank accession numbers). For *C. gibbsi* samples, an 810-bp fragment of the cyt *b* gene was amplified and sequenced using primers and conditions of Fluker, Pezold & Minton (2011). For *E. tallapoosae* samples, the complete cyt *b* gene (1140 bp) was amplified and sequenced using primers and conditions of Song, Near & Page (1998). All mtDNA sequences were read on an ABI PRISM® 3100 (Applied Biosystems) genetic analyzer (Steven Johnson Molecular Systematics Laboratory, University of Alabama) and aligned by eye.

Population genetic structure and differentiation: microsatellites and mtDNA

Population genetic structure, differentiation (i.e. F_{ST} for microsatellites and ϕ_{ST} for mtDNA) and associated significance were assessed by analysis of molecular variance (AMOVA; Excoffier, Smouse & Quattro, 1992) as implemented in ARLEQUIN with 10 000 permutations. We tested the null hypothesis that individuals from each tributary represent an arbitrary subsample relative to its respective group (reservoir or river). In these tests, tributaries OK, SA, MA and EL were treated as subpopulations of the larger reservoir group, and tributaries EM, SW, EG and CH were treated as subpopulations of the larger river group for both species, respectively. Because microsatellite DNA has a much higher mutation rate than mtDNA, the use of both markers in separate AMOVAs provides insight into historical (pre-impoundment) versus contemporary genetic structure (e.g. Emerson & Hewitt, 2005; Bos *et al.*, 2008; Anderson *et al.*, 2010). For example, if we observe concordant patterns of significant genetic structure from both markers in the reservoir group, then it is likely that genetic structuring was in place prior to reservoir construction (i.e. historical). However, if we find significant patterns of genetic structure in the reservoir group based on microsatellites, but not mtDNA, this may indicate that genetic structuring is contemporary, probably occurring after reservoir construction.

Population genetic structure and migration: microsatellite DNA

The program STRUCTURE 2.3.3 (Pritchard, Stephens & Donnelly, 2000) was used to determine the number of genetically differentiated clusters (*K*) within the reservoir and river groups, independently. This Bayesian clustering method uses allele frequency data to probabilistically assign individuals to groups without *a priori* population designations. If an individual's genotype reflects admixture, it is assigned to two or more groups, allowing an estimate of ancestry of an individual's genotype from one group or several groups (Pritchard *et al.*, 2000; Falush, Stephens & Pritchard, 2003). Four STRUCTURE analyses were conducted independently to determine *K* for reservoir and river groups of both species: (i) *C. gibbsi* from reservoir tributaries OK, SA, MA, EL, (ii) *C. gibbsi* from river tributaries EM, SW, EG, CH, (iii) *E. tallapoosae* from reservoir tributaries OK, SA, MA, EL and (iv) *E. tallapoosae* from river tributaries EM, SW, EG, CH. For each analysis, we conducted 10 replications for each *K* ranging from 1 to 6 (400 000 iterations with 100 000 burnin) under a model allowing admixture of genotypes and correlated allele frequencies between populations (Falush *et al.*, 2003). Appropriate *K* values for each analysis were evaluated by plotting the average 'estimated Ln probability of the data' (LnPD) for each *K* and by using the ad hoc summary statistic ΔK (Evanno, Regnaut & Goudet, 2005).

The possibility of isolation-by-distance among sampling locations was assessed for the reservoir and river groups independently using the Mantel permutation test

(Mantel, 1967). For Mantel tests, matrices of pairwise F_{ST} values were compared to geographical distance between sampling locations using 10 000 permutations in ARLEQUIN.

The program BAYESASS 1.3 (Wilson & Rannala, 2003) was used to infer the rate and direction of recent migration between tributaries within reservoir and river groups for both species, respectively. Using a Bayesian framework and a MCMC sampling scheme with multilocus genotype data, this method estimates the posterior probability of an individual's migration history. An important feature of BAYESASS is that conformity of populations to key demographic assumptions (e.g. HWE, mutation-drift equilibrium) is relaxed (Wilson & Rannala, 2003), providing greater confidence that migration rates reflect post-divergence gene flow, rather than ancestral population forces such as lineage sorting (Kane *et al.*, 2009; Scascitelli *et al.*, 2010). Although methodologies differ between STRUCTURE and BAYESASS, their combined use can provide complementary information about contemporary gene flow between populations (Kane & King, 2009).

Recent demographic history: microsatellite DNA

Two complementary methods were employed to assess whether populations have experienced recent declines (bottlenecks) in effective population size (N_e). The H_e excess test was implemented with the program BOTTLENECK 1.2.02 (Cornuet & Luikart, 1996). This method takes advantage of the observation that allele numbers are reduced faster than gene diversity in populations that have experienced recent reductions in N_e. Consequently, in recently bottlenecked populations, gene diversity (H_e) will be significantly greater than heterozygosity expected based on number of alleles and sample size (H_{eq}, Luikart & Cornuet, 1998; Luikart *et al.*, 1998; Piry, Luikart & Cornuet, 1999). Wilcoxon signed-rank tests were used to assess the significance of deviations between H_e and H_{eq}, implementing 10 000 replicates under a two-phase model (TPM), in which 95% of mutations were single-step with a 12% variance of multi-step mutations (Piry *et al.*, 1999). The second approach estimated the ratio of number of alleles to the range in allele size (M) and compared them to population specific critical M values (M_c) with the programs M_P_Val.exe and Critical_M.exe (Garza & Williamson, 2001). M-ratios were estimated for populations of both species under a TPM with 90% single-step mutations (p_s), average size of non-one-step mutations (Δ_g) = 3.5 and pre-bottleneck Θ values of 10 (N_e = 5000). To avoid inferences of bottlenecks where none are present, the settings above follow what is considered to be a conservative parameter set by Garza & Williamson (2001). We used 10 000 simulations to assess the probability that a smaller M-ratio would be expected under equilibrium conditions.

These two bottleneck methods test for decreases in population size due to different demographic processes. The H_e excess method is more likely to identify reductions in N_e that are very recent or less severe (Luikart *et al.*, 1998; Garza & Williamson, 2001; Williamson-Natesan, 2005). Alternatively, the M-ratio method is more powerful at detecting prolongued, severe or older reductions in N_e, even when populations may have had time to recover (Garza & Williamson, 2001; Williamson-Natesan, 2005).

Results

Compatibility of landscapes

Prior to construction of Martin Lake, the four streams of the reservoir group entered the Tallapoosa River within 2–23 km of one another (average distance between mouths 11.83 km). The four streams of the river group enter the Tallapoosa River within 2–19 km of one another (average distance between mouths 9.83 km). Distance between sampling sites ranged from 27 to 59 km (average = 44.17 km) among tributaries in the reservoir group and 3–53 km (average = 27.67 km) among tributaries in the river group. Comparisons of the distance between stream mouths ($Z = -0.48$, $P = 0.63$) and the distance between sampling sites ($Z = -1.62$, $P = 0.11$) were not significantly different between landscapes.

Characteristics of microsatellite loci

Of the 10 loci amplified for *C. gibbsi*, all were polymorphic and easily scored, and MICRO-CHECKER detected no typographical errors within the data set. However, Cylu_04_H06 suffered from potential scoring errors due to stutter and Can6EPA and Rhca24 exhibited an excess of homozygotes, indicating the possibility of null alleles. Following Bonferroni correction, 10 of the 80 tests for HWE showed significant deviations, all of which involved loci Cylu_04_H06 and Rhca24. Pairwise comparisons of LD across all loci revealed evidence for linkage between loci Rhca24 and Lco1, with no evidence for linkage among all other pairs. Due to the multiple deviations above, loci Cylu_04_H06 and Rhca24 were removed from the data set, and the remaining eight loci were used in subsequent analyses.

Eight of the 10 loci amplified for *E. tallapoosae* were polymorphic and easily scored, and MICRO-CHECKER detected no typographical or genotypic errors within the data set. Loci EosC6 and EosC112 were monomorphic across all samples. Following Bonferroni correction, 2 of the 57 tests for HWE showed significant deviations, both of which involved locus Esc57. No loci showed evidence of linkage. For the reasons above, loci EosC6, EosC112 and Esc57 were removed from the data set, and the remaining seven loci were used in subsequent analyses.

Genetic diversity: microsatellite loci

For *C. gibbsi*, reservoir tributaries showed lower levels of A (8.000–11.625) as compared to river tributaries (10.750–12.250), and values of AR were significantly lower in reservoir tributaries compared with river tributaries ($P < 0.05$) (Table 2). Both measures of heterozygosity (Table 2) showed substantial overlap between reservoir and river groups, with no significant differences. The number of private alleles was slightly higher within reservoir tributaries (7–16) as compared to within river tributaries (1–11) (Table 2).

For *E. tallapoosae*, no comparisons of genetic diversity (AR, H_o, H_e) were significantly different between groups. Reservoir tributaries showed similar levels of genetic diversity (A 7.667–13.286, H_e 0.689–0.754) as compared to river tributaries (A 8.833–11.000, H_e 0.645–0.714) (Table 3). The number of private alleles varied more within reservoir tributaries (3–28) than within river tributaries (4–10) (Table 3).

Genetic structure and differentiation

The microsatellite DNA-based AMOVA for *C. gibbsi* revealed low, but significant levels of genetic differentiation between tributaries within the reservoir group (average $F_{ST} = 0.05$, $P < 0.0001$; Table 4). Genetic differentiation between tributaries within the river group was lower than in the reservoir group, but significant (average $F_{ST} = 0.03$, $P < 0.0001$; Table 4). All pairwise F_{ST} estimates among tributaries within the reservoir group were highly significant ($P < 0.0001$ for all; see Table S1). Except for the SW-CH pair ($P = 0.704$), pairwise F_{ST} estimates among tributaries within the river group were low, but significant ($P < 0.01$; see Table S2). The Mantel tests for *C. gibbsi* revealed no significant association with genetic differentiation and geographical distance between sample sites within the reservoir group ($r = -0.768$, $P = 0.915$) or the river group ($r = -0.004$, $P = 0.589$). The mtDNA-based AMOVA for *C. gibbsi* from the reservoir group revealed low, but significant levels of genetic differentiation between tributaries (average $\phi_{ST} = 0.21$, $P = 0.016$; Table 4). After Bonferroni correction, however, there were no significant differences in pairwise comparisons of differentiation (see Table S3). The mtDNA-based AMOVA for *C. gibbsi* from the river group showed that genetic differentiation between tributaries was not different from zero (average $\phi_{ST} = 0.10$, $P = 0.087$; Table 4), and all pairwise comparisons of differentiation were non-significant (see Table S4).

The microsatellite DNA-based AMOVAs for *E. tallapoosae* revealed moderate levels of genetic differentiation

Table 2 Sample sizes, genetic diversity estimates (averaged over eight microsatellite loci) and results from H_e excess and *M*-ratio tests for *Cyprinella gibbsi*. Significant *P*-values ($P < 0.05$) are shown in bold. Symbols correspond to Fig. 1

Group/Tributary	Symbol	*N*	A	AR	PA	H_o	H_e	*P*-value (H_e excess)	Critical *M*	*M*-ratio	*P*-value (*M*)
Reservoir											
Oakachoy	OK	30	8.625	8.625	7.0	0.656	0.702	0.770	0.666	0.647	**0.027**
Sandy	SA	30	11.625	10.752	16.0	0.708	0.745	0.902	0.665	0.750	0.357
Manoy	MA	30	8.000	7.564	8.0	0.692	0.671	0.809	0.663	0.653	**0.030**
Elkahatchee	EL	30	9.625	9.221	12.0	0.724	0.780	0.125	0.668	0.676	0.069
Group mean	OK, SA, MA, EL	120	9.469	8.950	10.8	0.695	0.725	–	–	–	–
River											
Emuckfaw	EM	30	12.250	11.553	11.0	0.674	0.750	0.994	0.668	0.792	0.627
Sweetwater	SW	30	11.750	10.981	6.0	0.696	0.729	0.986	0.667	0.715	0.186
Eagle	EG	30	10.750	9.926	1.0	0.608	0.655	0.998	0.653	0.680	0.074
Chatahospee	CH	30	10.750	10.060	10.0	0.692	0.723	0.422	0.666	0.720	0.195
Group mean	EM, SW, EG, CH	120	11.375	10.630	7.0	0.667	0.714	–	–	–	–

N, number of individuals; A, mean number of alleles per locus; AR, allelic richness; PA, number of private alleles; H_o, heterozygosity observed; H_e, heterozygosity expected.

Table 3 Sample sizes, genetic diversity estimates (averaged over seven microsatellite loci) and results from H_e excess and *M*-ratio tests for *Etheostoma tallapoosae*. Significant *P*-values ($P < 0.05$) are shown in bold. Symbols correspond to Fig. 1

Group/Tributary	Symbol	*N*	A	AR	PA	H_o	H_e	*P*-value (H_e excess)	Critical *M*	*M*-ratio	*P*-value (*M*)
Reservoir											
Oakachoy	OK	29	7.833	6.716	3.0	0.671	0.639	1.000	0.649	0.599	**0.012**
Sandy	SA	30	13.287	12.986	28.0	0.713	0.738	0.406	0.662	0.829	0.805
Manoy	MA	30	9.833	8.352	10.0	0.755	0.754	0.422	0.653	0.577	**0.004**
Elkahatchee	EL	30	7.667	6.589	5.0	0.710	0.689	0.281	0.648	0.657	0.061
Group mean	OK, SA, MA, EL	119	9.655	8.661	11.5	0.712	0.705	–	–	–	–
River											
Emuckfaw	EM	30	10.333	8.799	10.0	0.626	0.645	0.961	0.648	0.604	**0.012**
Sweetwater	SW	30	11.000	9.262	9.0	0.700	0.675	0.945	0.652	0.653	0.056
Eagle	EG	30	11.000	9.389	9.0	0.706	0.714	0.945	0.649	0.615	**0.017**
Chatahospee	CH	30	8.833	7.565	4.0	0.620	0.659	0.922	0.650	0.668	0.080
Group mean	EM, SW, EG, CH	120	10.292	8.754	8.0	0.663	0.673	–	–	–	–

N, number of individuals; A, mean number of alleles per locus; AR, allelic richness; PA, number of private alleles; H_o, heterozygosity observed; H_e, heterozygosity expected.

Table 4 Analysis of molecular variance (AMOVA) for reservoir and river groupings of *Cyprinella gibbsi* and *Etheostoma tallapoosae* based on microsatellite DNA loci and mitochondrial (mt) DNA loci. Asterisks represent AMOVAs with non-significant results ($P > 0.05$) or with non-significance for all pairwise comparisons of fixation indices (F_{ST} and ϕ_{ST}) following Bonferroni correction (see Tables S1–S4 in Supporting information for all pairwise comparisons)

Species/Group/ Data type	Source of variation	d.f.	SS	% of variance	Fixation index	*P*-value
C. gibbsi						
Reservoir						
Microsatellites	Among tributaries	3	35.81	5.06	$F_{ST} = 0.05$	<0.0001
	Within tributaries	236	670.52	94.94	–	
mtDNA	Among tributaries	3	8.42	21.11	$\phi_{ST} = 0.21$	0.0016*
	Within tributaries	17	19.87	78.89	–	
River						
Microsatellites	Among tributaries	3	23.64	2.92	$F_{ST} = 0.03$	<0.0001
	Within tributaries	236	663.03	97.08	–	
mtDNA	Among tributaries	3	6.60	10.26	$\phi_{ST} = 0.10$	0.0867*
	Within tributaries	16	22.40	89.74	–	
E. tallapoosae						
Reservoir						
Microsatellites	Among tributaries	3	59.47	12.14	$F_{ST} = 0.12$	<0.0001
	Within tributaries	234	502.82	87.86	–	
mtDNA	Among tributaries	3	6.23	27.37	$\phi_{ST} = 0.27$	0.0116*
	Within tributaries	17	11.87	72.63	–	
River						
Microsatellites	Among tributaries	3	18.26	3.29	$F_{ST} = 0.03$	<0.0001
	Within tributaries	236	472.25	96.71	–	
mtDNA	Among tributaries	3	13.79	13.53	$\phi_{ST} = 0.14$	0.1007*
	Within tributaries	17	42.97	86.47	–	

d.f., degrees of freedom; SS, sum of squares.

between reservoir tributaries (average $F_{ST} = 0.12$, $P < 0.0001$), while values from the river group were substantially lower (average $F_{ST} = 0.03$, $P < 0.0001$), being approximately one-third that of the reservoir group (Table 4). All pairwise F_{ST} estimates among tributaries within their respective group were significant ($P < 0.01$ for all; see Tables S1, S2). The Mantel tests for *E. tallapoosae* revealed no significant association with genetic differentiation and geographical distance between sample sites within the reservoir group ($r = -0.084$,

$P = 0.622$) or the river group ($r = 0.403$, $P = 0.415$). The mtDNA-based AMOVA for *E. tallapoosae* from the reservoir group showed low, but significant levels of genetic differentiation between tributaries (average $\phi_{ST} = 0.27$, $P = 0.0116$; Table 4). After Bonferroni correction, however, there were no significant differences in pairwise comparisons of differentiation (see Table S3). The mtDNA-based AMOVA for *E. tallapoosae* from the river group showed that genetic differentiation between tributaries was not different from zero (average $\phi_{ST} = 0.14$, $P = 0.1007$; Table 4), and all pairwise comparisons of differentiation were non-significant (see Table S4).

The STRUCTURE runs for the *C. gibbsi* reservoir group showed that the mean LnPD rose to $K = 3$ and levelled off, corresponding to the plot of ΔK, indicating three genetically subdivided clusters (see Fig. S1). Group assignments for $K = 3$ indicated that OK and SA showed high (0.825) and moderately high (0.681) proportions of membership to cluster one, while MA and EL showed high proportions of membership to cluster two (0.925) and cluster three (0.833), respectively (Fig. 2c). Plots of the mean LnPD for the *C. gibbsi* river group revealed the highest values at $K = 1$ and decreased for subsequent K values (see Fig. S1). Although the plot of ΔK indicated $K = 2$ (not shown), estimates of $K = 1$ using the ΔK method are not possible (Evanno *et al.*, 2005). Further examination revealed symmetric proportions of membership of the four tributaries for all K values, confirming $K = 1$ for the *C. gibbsi* river group (Fig. 2d).

The STRUCTURE runs for the *E. tallapoosae* reservoir group showed that the mean LnPD rose to $K = 4$ and decreased thereafter, whereas the plot of ΔK indicated $K = 2$ (see Fig. S1). Evanno *et al.* (2005) showed that determination of K based solely on values of LnPD is insufficient, sometimes resulting in overestimates of K. However, when discrepancies arise between LnPD and ΔK, it is often best to test for substructure within the highest hierarchical groupings by analysing them separately (Evanno *et al.*, 2005). Thus, independent runs of our $K = 2$ groupings (OK + EL and SA + MA; Fig. 2e) were conducted in STRUCTURE with K values ranging from 1 to 4 and the same settings listed above. Analysis of each group indicated that each tributary could be assigned to a distinct cluster with high probability ($K = 2$ for each analysis; not shown), suggesting $K = 4$ for the reservoir group of *E. tallapoosae*. Group assignments for $K = 4$ indicated that each tributary had high average proportions of membership to the STRUCTURE groupings (OK = 0.942, SA = 0.952, MA = 0.918 and EL = 0.952; Fig. 2f). Plots of the mean LnPD for the *E. tallapoosae* river group revealed the highest values at $K = 1$ and decreased for subsequent K values (see Fig. S1). Further examination revealed symmetric proportions of membership of the four tributaries for all K values, confirming that $K = 1$ for the *E. tallapoosae* river group (Fig. 2g).

Recent migration

Estimates of contemporary migration for the *C. gibbsi* reservoir group were in agreement with STRUCTURE inferred clusters. The results indicated one-way migration [0.288 (0.212–0.325)] from OK into SA, while all other comparisons revealed a high proportion of individuals derived from their source tributary (Table 5). Migration was much higher for *C. gibbsi* between river tributaries, with strong signal of recent upstream migration from EM into all other river tributaries (Table 6). The analysis revealed possible downstream migration from SW into EM [0.130 (0.008–0.254)], although the large 95% confidence intervals suggest limited support for this finding (Table 6).

Estimates of recent migration for the *E. tallapoosae* reservoir group revealed moderate levels of reciprocal migration between OK and EL (Table 5), which was partially in agreement with STRUCTURE results that indicated $K = 2$ (Fig. 2e). However, SA and MA showed a lack of recent migration as evidenced by the high proportion of individuals derived from their source tributaries, respectively (Table 5). Migration estimates among tributaries within the *E. tallapoosae* river group indicated greater connectivity as compared to tributaries within the

Table 5 Migration rates among reservoir tributaries for *Cyprinella gibbsi* and *Etheostoma tallapoosae* depicted by mean values of the posterior distributions of m (the migration rate into each tributary). Columns represent tributaries of migrant origin and rows represent tributaries receiving migrants. Proportions of individuals derived from the source tributary each generation are listed along the diagonal. Migration rates ≥0.10 are in bold. Symbols correspond to Fig. 1.

Species/Tributary	OK	SA	MA	EL
C. gibbsi				
Oakachoy	**0.986**	0.004	0.005	0.005
Sandy	**0.288**	**0.679**	0.017	0.016
Manoy	0.005	0.004	**0.986**	0.004
Elkahatchee	0.024	0.007	0.007	**0.961**
E. tallapoosae				
Oakachoy	**0.677**	0.008	0.007	**0.308**
Sandy	0.009	**0.980**	0.006	0.005
Manoy	0.006	0.004	**0.984**	0.006
Elkahatchee	**0.308**	0.007	0.007	**0.677**

OK, Oakachoy Creek; SA, Sandy Creek; MA, Manoy Creek; EL, Elkahatchee Creek.

Table 6 Migration rates among river tributaries for *Cyprinella gibbsi* and *Etheostoma tallapoosae* depicted by mean values of the posterior distributions of m (the migration rate into each tributary). Columns represent tributaries of migrant origin and rows represent tributaries receiving migrants. Proportions of individuals derived from the source tributary each generation are listed along the diagonal. Migration rates ≥0.10 are in bold. Symbols correspond to Fig. 1.

Species/Tributary	EM	SW	EG	CH
C. gibbsi				
Emuckfaw	**0.837**	**0.130**	0.014	0.019
Sweetwater	**0.301**	**0.679**	0.009	0.010
Eagle	**0.306**	0.008	**0.678**	0.008
Chatahospee	**0.294**	0.016	0.009	**0.681**
E. tallapoosae				
Emuckfaw	**0.677**	**0.298**	0.008	0.017
Sweetwater	0.010	**0.909**	0.036	0.045
Eagle	0.008	**0.292**	**0.686**	0.015
Chatahospee	0.004	0.008	0.004	**0.984**

EM, Emuckfaw Creek; SW, Sweetwater Creek; EG, Eagle Creek; CH, Chatahospee Creek.

reservoir group (Table 6). There was evidence for up- and downstream migration from SW into EM and EG [0.298 (0.232–0.328) and 0.292 (0.207–0.329), respectively], while CH showed relatively little migration compared with other river tributaries (Table 6).

Demographic history

Results from both tests of population bottlenecks (H_e excess and M-ratio) are presented for each tributary and for each group (reservoir and river) in Tables 2 and 3 for *C. gibbsi* and *E. tallapoosae*, respectively. The H_e excess analyses indicated no evidence for significant excess of heterozygotes under the TPM for any of the groupings for either species. Alternatively, M values for *C. gibbsi* from reservoir tributaries OK and MA were significantly lower than their respective M_c values, while none of the values from the river tributaries showed significant deviations (Table 2). For *E. tallapoosae*, two tributaries within the reservoir group (OK and MA) and two tributaries within the river group (EM and EG) showed significantly lower M values, as compared to their respective M_c values (Table 3).

Discussion

Understanding how species are affected by human-induced fragmentation is a crucial first step in predicting their persistence in altered settings. For both species of stream fishes examined here, we found genetic discontinuities between neighbouring tributaries that have been fragmented by reservoir construction. This finding was in stark contrast to the high degree of continuity for both species between reference tributaries in a natural river setting. Measures of genetic diversity and migration were differentially affected in the two species, indicating that stream fishes with relatively high dispersal abilities may be equally or more susceptible to reservoir fragmentation as compared to species with putatively low dispersal abilities. These data revealed that reservoir fragmentation influences genetic patterns and processes in stream fishes, which is of particular interest in biodiversity hotspots such as the south-eastern United States where hydroelectric and recreational reservoirs restrict connectivity in aquatic systems.

Genetic structure and recent migration

Our hypothesis that genetic structure should be greater among neighbouring tributaries isolated by Martin Lake (reservoir group) relative to neighbouring tributaries in an unimpounded setting (river group) was supported for both species. The STRUCTURE analyses for *C. gibbsi* revealed three genetically distinct clusters within the reservoir group, while the river group was resolved as a single homogenous cluster. Although levels of differentiation were low among reservoir tributaries, all pairwise comparisons of differentiation were significant and average F_{ST} values were nearly twice those observed among river tributaries. Similarly, STRUCTURE analyses of *E. tallapoosae* recovered at least two genetically distinct clusters, with strong support for four unique clusters within the reservoir group. Alternatively, the river group was a single homogenous cluster. Levels of genetic differentiation (F_{ST}) among reservoir tributaries were approximately three times the values observed among river tributaries. Genetic differentiation was not associated with geographical distance between sampling sites within groups for either species examined. For the reservoir group, this provides additional evidence that patterns of genetic structure have been shaped by the isolating effects of the reservoir, despite the geographical distance between sampling locations.

The finding of recent or ongoing gene flow between some of the reservoir tributaries was surprising for both species, given their strict habitat requirements of flowing, small- to medium-sized streams. Within the *C. gibbsi* reservoir group, there was evidence for one-way migration from Oakachoy Creek into Sandy Creek, suggesting recent or ongoing gene flow between the two southernmost eastern and western reservoir tributaries. Within the *E. tallapoosae* reservoir group, there was evidence for

reciprocal migration between Oakachoy Creek and Elkahatchee Creek, suggesting recent or ongoing gene flow between the two western reservoir tributaries. There are several possible explanations for these two examples of trans-reservoir migration by both species. The first is that these fishes are able to traverse the inhospitable habitat of Martin Lake. This scenario is unlikely given their strict habitat requirements, lack of collections from the reservoir (Howell & Williams, 1971) and the high levels of genetic structure between the other reservoir tributaries. A second explanation is that some reservoir tributaries share gene flow via migration along the Martin Lake shoreline. This scenario could be accomplished through gene flow in stepping-stone fashion between neighbouring streams, where individuals occasionally navigate short stretches of reservoir shoreline between streams. This scenario is possible, but seemingly unlikely given that the distances between appropriate habitat is extensive between neighbouring reservoir tributaries. For example, even if neighbouring tributary mouths are relatively close to one another, much of the lower stretches of the tributaries flowing into the reservoir are impounded and represent inhospitable habitat. A third possibility is that the reservoir impedes gene flow between tributaries, but shared ancestral allelic variation results in the false identification of gene flow and connectivity between tributaries. For example, alleles shared by individuals among tributaries of the reservoir group during pre-impoundment times may have been retained by individuals in some neighbouring tributaries, but lost in others upon isolation by the formation of the reservoir. This phenomenon is often observed in recently isolated species or populations with relatively large population sizes (Pamilo & Nei, 1988). Although we cannot confidently distinguish between occasional shoreline migration and shared ancestral variation, the patterns of genetic structure observed in both species occupying reservoir tributaries suggest the latter. It would be intuitive to assume that occasional shoreline migration would be more common between geographically proximate tributaries, which is not the case for our system. For example, recent gene flow would be expected between Sandy and Manoy creeks or between Manoy and Elkahatchee creeks, but neither species showed evidence for recent connectivity between these tributaries.

Historical versus contemporary genetic structure in the reservoir group

Determining whether genetic structure in a fragmented landscape is the product of contemporary fragmentation, or simply historical factors, is crucial in any landscape genetic study (Anderson *et al.*, 2010; Storfer *et al.*, 2010). Because of differences in sampling strategy, it is difficult to make direct comparisons of our results with previously published studies of genetic structure within *C. gibbsi* and *E. tallapoosae*. However, a few conflicting patterns warrant further discussion. For example, Connelly *et al.* (2006) examined mtDNA variation throughout most of the range of *C. gibbsi*, including samples from Oakachoy Creek, two tributaries that enter Martin Lake near its upstream terminus (upstream of Elkahatchee Creek), and seven sites upstream of Martin Lake (all above Chatahospee Creek) encompassing the upper extent of the Tallapoosa River drainage. Their results suggested significant genetic structuring among four groups; each of the three tributaries entering Martin Lake was distinct, and the remaining seven sites spanning the upper Tallapoosa River drainage were resolved in a single distinct group. However, each of the three Martin Lake populations of *C. gibbsi* defined by Connelly *et al.* (2006) shares haplotypes with populations in the uppermost reaches of the Tallapoosa River, suggesting somewhat recent associations of upper and lower Tallapoosa River populations of *C. gibbsi*. Brogdon *et al.* (2003) found a similar pattern of mtDNA structure for *E. tallapoosae*. However, they only analysed two of the sites of Connelly *et al.* (2006) that enter Martin Lake above Elkahatchee Creek, and 11 sites in the upper Tallapoosa River drainage. Brogdon *et al.* (2003) suggested the possibility that genetic structure of *E. tallapoosae* (1.04% mtDNA sequence divergence) between the two tributaries entering Martin Lake was not due to reservoir isolation, but more likely to bottleneck colonisation events from an ancestral polymorphic population in the southern portion of the Tallapoosa River. Connelly *et al.* (2006) further suggested 'the main channel of the southern portion of the Tallapoosa River (historically) and Martin Lake (over the last 76 years) has served as a barrier to migration in both species' (*C. gibbsi* and *E. tallapoosae*). While both Brogdon *et al.* (2003) and Connelly *et al.* (2006) suggested that genetic structure in both species in the Martin Lake area has a historical basis (i.e. pre-impoundment isolation), we found no real evidence of historical influences on genetic structure in either species across a fragmented and a non-fragmented landscape. We believe that the conflicting results are due to their spatial scale and sampling strategy, which differed widely from ours. In particular, Brogdon *et al.* (2003) and Connelly *et al.* (2006) had relatively few sites in the Lower Tallapoosa River (i.e. Martin Lake tributaries) and numerous sites in the upper Tallapoosa River,

whereas our study focused on two differing landscapes that were equally sampled on a relatively equal spatial scale.

The combined use of microsatellite loci and mtDNA in our study further allows us to account for the temporal scale at which genetic change is expected to occur. For example, mtDNA evolves at a comparatively slower rate than microsatellite DNA loci and is best suited to evaluate change over historical timescales (Anderson *et al.*, 2010). Thus, the lack of significant mtDNA structure in the reservoir group for both species in our study suggests a lack of historical (i.e. pre-impoundment) genetic structure between tributaries. Two limitations deserve additional comment with respect to our analysis of mtDNA. First, our assumption of a slower evolutionary rate for mtDNA versus microsatellites holds true for mutation, but genetic drift in the short term can lead to more rapid fixation of mtDNA alleles and greater genetic structure because effective population size of mtDNA is approximately four times smaller compared with nuclear DNA (Ballard & Whitlock, 2004). However, we believe that the use of mtDNA in our study was appropriate for explicitly testing for historical signatures of genetic structure (i.e. pre-impoundment). If there were high levels of mtDNA haplotype divergence between reservoir tributaries prior to the construction of the reservoir, then this pattern should still be detectable. Even if rapid drift shifted allele frequencies in mtDNA since isolation from the reservoir, the degree of differences between mtDNA haplotypes would have remained relatively the same over this time period. Second, the relatively small sample sizes used for mtDNA in our AMOVAs (i.e. 20–21 per group, 5–6 per subpopulation) could potentially lead to estimates of non-significant genetic structure. Although haplotype divergence is small between sample sites in the reservoir group, further supporting a lack of historic structure, larger sample sizes may resolve significant genetic structure based on mtDNA. Alternatively, the finding of significant microsatellite DNA-based structure in the reservoir group indicates a contemporary process. It is important to note that the degree of population structure based on microsatellite DNA data is not beyond the realm of expectations due to habitat fragmentation by Martin Lake. Studies of closely related species suggest both *C. gibbsi* and *E. tallapoosae* exhibit generation times of 1–4 years (Outten, 1958; Pflieger, 1975; Page, 1983; Ferguson, 1990), indicating that both species have been fragmented in tributaries of Martin Lake for about 21–84 generations. Simulation studies by Murphy *et al.* (2008) showed that landscape effects on genetic structure can be detectable in as few as five generations for short-lived species. Several empirical studies of terrestrial vertebrates support this finding and showed that genetic structure between populations fragmented within the last 100 years (e.g. Riley *et al.*, 2006; Zellmer & Knowles, 2009; Tracy & Jamieson, 2011). Collectively, the results from this study and a recent body of work on similar stream-inhabiting fishes (i.e. Skalski *et al.*, 2008; Franssen, 2011; Hudman & Gido, 2013) suggest that reservoir-induced fragmentation can affect gene flow and genetic structure in stream fishes over relatively brief timescales.

Genetic diversity and recent demographic history

Because the reservoir tributaries in our study are putatively isolated from one another and have lost a substantial proportion of their former habitat (stream length), we hypothesised that genetic diversity would be lower for both species in the fragmented reservoir group as compared to the river group. Our data supported this hypothesis for *C. gibbsi*, but not for *E. tallapoosae*. The reservoir group of *C. gibbsi* showed significantly lower allelic richness when compared to the river group. Conversely, measures of genetic diversity were not significantly different between reservoir and river groups of *E. tallapoosae*.

Although we found no evidence of very recent or minor bottlenecks using the H_e excess method, the differences in genetic diversity between reservoir and river groups of *C. gibbsi* were corroborated by more prolongued or severe bottlenecks revealed by *M*-ratio tests. For *C. gibbsi*, when we analysed *M*-ratios of each tributary within a group, Oakachoy and Manoy creeks were significantly below their critical values, and Elkahatchee Creek, while not significant, showed a value low enough to suggest genetic bottleneck ($M < 0.68$, Garza & Williamson, 2001). In comparison, no significant reductions in *M* were observed for *C. gibbsi* in river group, with only Eagle Creek showing a relatively low *M* value. For *E. tallapoosae*, bottlenecks were strongly supported by *M* values in at least two reservoir (Oakachoy and Manoy creeks) and river (Emuckfaw and Eagle creeks) tributaries, respectively. Additional evidence from low, but non-significant, *M*-ratios was evident in Elkahatchee, Sweetwater and Chatahospee creeks. These results suggest that *E. tallapoosae* from all reservoir and river tributaries in our study, excluding Sandy Creek, have experienced reductions in effective population size. The results for *E. tallapoosae* suggest that local bottlenecks have acted similarly on reservoir and river groups, which may be associated with the low dispersal ability of the species.

Interestingly, our results suggest that the amount of remaining habitat in reservoir streams affected genetic characteristics in both species. For example, there was strong signal of bottlenecks in both species within the reservoir tributary Manoy Creek, which has the smallest amount of remaining stream habitat. Additionally, both species exhibit notably higher levels of allelic variation within reservoir tributary Sandy Creek, which has the largest remaining proportion of stream habitat compared with other reservoir tributaries sampled.

Dispersal ability and reservoir effects

Given the relatively higher dispersal ability of *C. gibbsi*, we hypothesised that reservoir effects (i.e. increased genetic structure, decreased genetic diversity) would be greater in the relatively sessile *E. tallapoosae*. One aspect of the results that supported our hypothesis was the greater degree of genetic differentiation between reservoir tributaries (average $F_{ST} = 0.12$), relative to differentiation between river tributaries (average $F_{ST} = 0.03$) for *E. tallapoosae*. The levels of average differentiation between reservoir and river groups of *C. gibbsi* were not nearly as pronounced. Further, the STRUCTURE analyses revealed a greater number of genetically distinct clusters in the *E. tallapoosae* reservoir group compared with *C. gibbsi*. Nevertheless, our data did not fully support this hypothesis and revealed that reservoirs may affect species with relatively high dispersal abilities equally or more strongly compared with species with relatively low dispersal abilities. For example, allelic diversity was significantly reduced in the reservoir group of *C. gibbsi*, where genetic bottlenecks were more prevalent compared with the river group. The same comparisons between *E. tallapoosae* groups revealed no such reservoir effect on genetic diversity, where reservoir and river groups have both probably suffered from bottlenecks. Published data on population size are lacking for the species used in this study, but we suspect that population size and home range size play an important part in the differing patterns of genetic diversity. For example, the mid-water dwelling *C. gibbsi* may require extended spatial scales to maintain adequate levels of gene flow and stable population sizes, which may explain why reservoir-induced isolation has affected genetic diversity in the reservoir group for this species, but not *E. tallapoosae*. It has also been suggested that close relatives of *E. tallapoosae* (subgenus *Ulocentra*) can maintain high densities and large effective population sizes, despite structuring on fine geographical scales (Porter *et al.*, 2002; Powers & Mayden, 2007). Future studies incorporating species density and population size may help to clarify the relationship between dispersal ability, population size, home range size and reservoir impacts to genetic diversity.

Conservation implications

Our study has revealed that fragmentation of stream connectivity by Martin Lake has negatively affected genetic characteristics of two stream fishes in the Tallapoosa River. A major concern for species or populations that are fragmented by landscape alteration is connectivity via gene flow (Fischer & Lindenmayer, 2007). The reductions in gene flow between reservoir-fragmented populations in this study increase the risk of local population extinction and can lead to genetic problems such as inbreeding, which collectively influence population persistence (Lande, 1987; Spielman, Brook & Frankham, 2004; Allendorf & Luikart, 2007). Consistent with recent studies of stream fishes in fragmented river systems (e.g. Skalski *et al.*, 2008; Franssen, 2011; Hudman & Gido, 2013), our study showed that small-stream species can experience genetic impacts from reservoir fragmentation over a relatively short time period (approximately 85 years). Particularly alarming to us is that both of the species in this study have experienced adverse genetic impacts from reservoir-induced fragmentation despite being widespread and common throughout the Tallapoosa River system (Boschung & Mayden, 2004). Thus, it appears that reservoir-induced fragmentation may have a greater impact on 'common' stream species than previously appreciated.

Our study also contributes to the growing body of work on the negative impacts of riverine impoundment on the freshwater fauna of the Mobile River basin (e.g. Williams, Fuller & Grace, 1992; Burke & Ramsey, 1995; Lydeard & Mayden, 1995; Pringle *et al.*, 2000). While our findings are particularly relevant to conservation strategies for small-stream inhabitants in the Mobile basin, we believe that our results and the results of others solidify the Mobile basin as a model system to better understand riverine impoundment and its impacts to organisms in freshwater systems worldwide.

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Supporting Information

Additional Supporting Information may be found in the online version of this article:

Appendix S1 Source of *Cyprinella gibbsi* and *Etheostoma tallapoosae* samples sequenced for the mitochondrial (mt) DNA cytochrome *b* gene.

Figure S1 Results from the STRUCTURE analyses, showing plots of the average Ln Probability of the data (LnPD) against number of STRUCTURE defined clusters (*K*), and rate of change in the LnPD between successive *K* values (ΔK; Evanno *et al.*, 2005) against *K*.

Table S1 Pairwise differentiation estimates (F_{ST}) from the microsatellite DNA-based AMOVA between reservoir tributaries for *Cyprinella gibbsi* (above diagonal) and *Etheostoma tallapoosae* (below diagonal).

Table S2 Pairwise differentiation estimates (F_{ST}) from the microsatellite DNA-based AMOVA between river tributaries for *Cyprinella gibbsi* (above diagonal) and *Etheostoma tallapoosae* (below diagonal).

Table S3 Pairwise differentiation estimates (ϕ_{ST}) from the mitochondrial DNA-based AMOVA between reservoir tributaries for *Cyprinella gibbsi* (above diagonal) and *Etheostoma tallapoosae* (below diagonal).

Table S4 Pairwise differentiation estimates (ϕ_{ST}) from the mitochondrial DNA-based AMOVA between river tributaries for *Cyprinella gibbsi* (above diagonal) and *Etheostoma tallapoosae* (below diagonal).

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